is taking. Along these lines, managers who specialize and hold more concentrated portfolios tend to have better performance.

Even with all these measures, finding outperforming mutual fund managers is not easy.

3.4. FLOWS

Mutual fund investors, in the words of Frazzini and Lamont (2008), are "dumb money." Cash pours into mutual funds when past returns have been high, but it tends to flow into the wrong funds. Subsequent returns end up being low. Friesen and Sapp (2007) estimate that dumb money flows—investors placing money in funds with high past returns but low future returns—costs investors 1.5% per year. Mutual fund companies trumpet the past high returns of funds in their advertising. This is selective, of course, because only funds with recent good performance get star billing. Money comes in, and future returns are disappointingly low.[36]

Mutual fund money is sticky. Money flows into good performing funds, but money tends not to leave bad performing funds at the same rate.[37] In jargon: the *flow-performance relationship is convex*. Figure 16.5 shows the relationship between flows into active funds over the following year (y-axis) as a function of the past returns of mutual funds (x-axis). At the end of each year from 1980 to 2012, funds are ranked into deciles by their performance over the previous year. The flows into those funds are recorded over the following year. The figure reports the weighted average next-year flows as a function of weighted average past-year returns, in both cases using AUM weights within each decile at the end of each year. In Figure 16.5, the slope of the flow relationship is less steep for negative past returns, while high past returns induce large increases in new money. Thus, investors do not punish losses in the same way they reward success. This investor inertia saved Janus.

3.5. BERK AND GREEN (2004) MODEL

Jonathan Berk, a finance professor at Stanford University, and Richard Green, my sometime co-author and a finance professor at Carnegie Mellon University, wrote

[36] For an academic study on mutual fund advertising, see Jain and Wu (2000). Gruber (1996) and Zheng (1999) show that mutual fund flows predict fund returns, at least in the short run. This is consistent with the Berk and Green (2004) argument (see below) that some investors can identify skilled managers, but performance in the long run suffers as the funds grow. Lou (2012) shows that mutual fund flows predict stock returns.

[37] See Ippolito (1992), Chevalier and Ellison (1997), Goetzmann and Peles (1997), and Sirri and Tufano (1998). We observe the same convex flow-performance relationship in hedge funds (see Agarwal, Daniel, and Naik 2004 and Ding et al. 2009) and pension funds (see Del Guercio and Tkac 2002). Bergstresser and Poterba (2002) find that new money (inflows) is sensitive to past performance, while withdrawals (outflows) have no statistically significant relation to past returns.